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MILITARY SPECIFICATION

RECORDER SET, SOUND, AN/UNH-6

This specification has been approved by the
Bureau of Aeronautics, Department of the Navy.

1. SCOPE

1.1 Scope - The equipment covered by this specification shall fulfill the requirements for a lightweight airborne magnetic tape recorder. The recorder is a high fidelity dual track type and is required to be lightweight, small in size, rugged, and versatile. The recorded tapes derived from this equipment shall be capable of play back on Recorder-Reproducer Set, Sound AN/UNQ-7, or equivalent.

1.2 Classification - The Recorder Set, Sound AN/UNH-6 shall be of one type and shall consist of the following:

Units	Type Designation	Max. Allow. Size ((nches) W	H	L/D	Max. Allow. Wt. (Lbs)	Appl. Para.
Recorder, Sound	RO-28/UNH-6	4-7/8	7-5/8	21-5/8	19.0	3.4.1
Control, Recorder	C-(*)/UNH-6	5-3/4	2-5/8	4-1/2	2.7	3.4.2
Indicator	IP-(*)/UNH-6	5-3/4	3	6-1/2	4.0	3.4.3
Mountings						
Mounting	MT-(*)/UNH-6	6-1/4	9	23-7/16	4.0	3.4.1.2

NOTE: The parentheses (*), in the type designation throughout this specification, shall be replaced by either a number or letter furnished by the Bureau of Aeronautics upon application by the contractor for assignment of nomenclature. The complete type number shall be used on nameplates, shipping records, and instruction books, as applicable.

1.3 Accessories

1.3.1 Equipment - The equipment shall be capable of operating with the following accessories which shall be supplied as part of this equipment.

1 reel - 7 inches in diameter and loaded with 1800 feet of mylar base tape

1 reel - 7 inches in diameter and unloaded.

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1.3.2 Other - When installed in aircraft, this equipment will be connected to and used with other equipments generally similar to those listed below. These other equipments are not a part of the AN/UNH-6 Sound Recorder Set and are not to be furnished therewith.

Microphone - NAF - 213264-6 or ANB-M-C1
or
Interphone - AN/AIC-4 or AN/AIC-14 or equivalent

2. APPLICABLE DOCUMENTS

2.1 General - The following documents of the issue in effect on the date of invitation for bids, form a part of this specification.

SPECIFICATIONS

Military

MIL-C-172	Cases; Bases, Mounting; and Mounts, Vibration (For Use with Electronic Equipment in Aircraft)
MIL-E-4682	Electron Tubes, Choise and Application of
MIL-W-5088	Wiring, Aircraft, Installation of
MIL-E-5400	Electronic Equipment, Aircraft, General Specification for
MIL-T-5422	Testing; Environmental, Aircraft Electronic Equipment
MIL-I-6181	Interference Limits, Tests and Design Requirements, Aircraft Electrical and Electronic Equipment
MIL-C-6781	Control Panel; Aircraft Equipment Rack-or Console-Mounted
MIL-P-7788	Plate, Plastic, Cockpit and Interior Controls Lighting
MIL-E-7894	Electric Power, Aircraft Characteristics of
MIL-P-17555	Preparation for Delivery of Electronic Equipment; Miscellaneous Electrical Equipment (Except Rotating Electrical Equipment) and Associated Repair Parts
MIL-T-18303	Test Procedures, Preproduction and Inspection for Aircraft Electronic Equipment; Format for
MIL-N-18307	Nomenclature and Nameplates for Airborne Electronic and Associated Equipment
MIL-F-25173	Fastener, Control Panel, Aircraft Equipment
MIL-R-21091	Recorder-Reproducer Set, Sound AN/UNQ-7

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Bureau of Aeronautics

EI-207	Control Panels; Electrical and Electronic Equipment, Rack or Console Mounted, Installation of
SAR-300	Engineering Change Proposals, Electronic Material Changes and Electronic Material Bulletins; Preparation of

STANDARDS

MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Air Force - Navy Aeronautical

MS 25245	Connections - Input Power Airborne Electronic Equipment; 28-Volt DC, 115-Volt Single Phase AC, and/or 115/200-Volt Three-Phase AC, or combinations

2.2 Availability of Documents

(1) Give both the title and number when requesting specifications, standards, drawings, and publications. All requests should be made via the cognizant Inspector of Naval Material or Bureau of Aeronautics Representative.

(2) Copies of this specification and applicable documents may be obtained upon application to the Commanding Officer, Naval Aviation Supply Depot, Philadelphia 11, Pennsylvania, Attention: Code ODPT.

(3) Copies of applicable drawings may be obtained upon application to the Bureau of Aeronautics, Navy Department, Washington 25, D. C., Attention: Avionics Division.

2.3 Precedence - When the requirements of the contract, this specification, or applicable subsidiary specifications are in conflict, the following precedence shall apply:

(1) Contract - The contract shall have precedence over any specification.

(2) This Specification - This specification shall have precedence over all applicable subsidiary specifications. Any deviation from this specification, or from subsidiary specifications where applicable, shall be approved in writing by the Bureau of Aeronautics.

(3) Referenced Specifications - Any referenced specification shall have precedence over all applicable subsidiary specifications referenced therein. All referenced specifications shall apply to the extent specified.

3. REQUIREMENTS

3.1 Parts and Materials - In the selection of parts and materials, fulfillment of major design objectives shall be the prime consideration. In so doing the following shall govern:

(1) Parts and materials as approved by Specification MIL-E-5400 shall be given first consideration.

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(2) When the characteristics of standard MIL parts and materials are such that their use will not fulfill the requirements because of size, weight, performance or other reasons, materials and parts shall be used which most nearly meet or exceed the requirements of the respective standard parts and material specifications.

(3) When a non-standard part is used, the meeting of general material requirements such as non-inflammability, fungus resistance, non-toxicity, etc., and the meeting of environmental conditions as required by Specification MIL-E-5400 or MIL-T-5422 shall be of prime consideration.

(4) In the selection of parts, the above considerations shall apply if the assembly is to built for detail maintenance. In the event the entire assembly is built to be expended, the assembly shall be constructed and the quality of the individual parts shall be such that the unit as a whole will pass material, electrical and environmental tests. Any assembly built to be expended shall be approved first by the Bureau of Aeronautics.

3.1.1 Non-Standard Parts and Material Approval - Approval for the use of non-standard parts and materials shall be obtained as outlined in Specification MIL-E-5400 except as follows:

(1) Mechanical parts shall be submitted as non-standard parts only when the material requirements of Specification MIL-E-5400 are not met.

(2) Non-standard parts of an electrical nature as defined in Specification MIL-E-5400, shall be approved before the preproduction model of the equipment is submitted for tests. The request for approval shall be forwarded to the Bureau of Aeronautics before the part is purchased or fabricated. In addition to or in lieu of the Non-Standard Part Data Sheet, the contractor shall submit an itemized list of the parts, their sources, and the contractor's drawing or purchase specification covering the parts.

3.1.2 Tube Types - The choise and application of electron tubes including transistors, used in the equipment shall be in accordance with the following requirements and in the sequence noted:

(1) MIL-E-4682

(2) The use of any transistor or tube not meeting the requirements of (1) above shall be approved by the Bureau of Aeronautics; however, before consideration for approval, the contractor shall submit the following information:

(a) Quantitative data on the electrical advantages and mechanical adequacy of the proposed transistor or tube over the most nearly equivalent type on the acceptable listings.

(b) Test data and performance limits of the transistor or tube which must be maintained to insure satisfactory and reliable performance in the equipment.

(c) Information as to whether the transistor or tube manufacturer has received Military or Navy qualification approval for the particular transistor or tube.

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3.1.2.1 Electron Tube Complement - A tube complement report shall be prepared in accordance with Specification MIL-E-4682 and submitted to the Bureau of Aeronautics for review and approval prior to preproduction testing. The tube complement shall include all electron tube types (including crystal diodes and transistors) used in the equipment.

3.1.3 Performance Objectives - Minimum size and weight, simplicity of operation, ease of maintenance, and an improvement in the performance and reliability of the specific functions beyond the requirements of this specification are objectives which shall be considered in the production of this equipment. Where it appears a substantial reduction in size and weight or improvement in simplicity of design, performance, ease of maintenance or reliability will result from the use of materials, parts and processes other than those specified in Specification MIL-E-5400, it is desired their use be investigated. When investigation shows advantages can be realized a request for approval shall be submitted to the Bureau of Aeronautics for consideration. Each request shall be accompanied by complete supporting information.

3.2 Design and Construction:

3.2.1 General - The equipment shall fulfill all applicable design and construction requirements of Specification MIL-E-5400 unless otherwise specified herein.

3.2.2 Service Conditions - The equipment shall operate satisfactorily under any of the environmental and mechanical service conditions or reasonable combination of these conditions as specified in Specification MIL-E-5400 for Class I equipment unless otherwise specified herein.

3.2.2.1 Equipment Attitude - The equipment shall meet all performance requirements when mounted on a surface which may be displaced from the normal horizontal plane through an angle of 30° to the horizontal. Operation at displacement angles in excess at 30° shall not cause damage to the equipment.

3.2.3 Standard Conditions - The following conditions shall be used as a basis to establish normal performance requirements and for making laboratory bench tests on the equipment.

Temperature	Room ambient (30° ± 10°C)
Altitude	Normal ground
Vibration	None
Humidity	Room ambient up to 90% relative humidity
Input Power Voltage	115 ± 1.0 V AC and/or 27.5 ± 0.5 V DC

3.2.4 Total Weight - The total weight of the equipment, excluding cables, shall be a minimum consistent with good design and shall not exceed 29.7 pounds.

3.2.5 Operational Stability - The equipment shall operate with optimum performance for 500 hours, continuously or intermittently, without the necessity for readjustment of any controls which are inaccessible during flight.

3.2.6 Operating Life

3.2.6.1 Reliable Operating Life - The overall equipment, except the capstan drive motor, shall have a reliable operating life of at least 2,000 hours without removal for bench servicing. Parts requiring servicing or replacement at the end of this interval to renew this service life shall be specified by the manufacturer. The same requirements shall apply to the capstan drive motor insofar as is practicable but in no case shall the reliable operating life thereof be less than 500 hours.

3.2.6.2 Total Operating Life - The equipment shall have a minimum total operating life of 10,000 hours with reasonable servicing and replacement of parts. Parts requiring replacement within this interval and the life of such parts shall be specified by the manufacturer.

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3.2.7 Interconnection Cabling - This equipment shall be capable of satisfactory operation using external wiring in accordance with the applicable requirements of Specification MIL-W-5088. The external wiring shall be unshielded, except that a minimum number of the individual wires may be shielded when demonstrated as necessary to meet interference control requirements and provided the assembly of the cable to its plugs may be easily accomplished.

3.2.8 Cables and Connectors - The equipment shall provide for the use of cables and connectors in accordance with Specification MIL-E-5400. However, cables and that portion of the connectors which is to be attached to the cables shall not be supplied as part of the equipment.

3.2.9 Console Controls - All remote control units shall be satisfactory for mounting in a console arrangement when used in aircraft. These units shall meet the requirements of Specification MIL-C-6781, shall have a plastic lighting plate conforming to Specification MIL-P-7788 and shall be equipped with fasteners conforming to Specification MIL-F-25173 for mounting in accordance with EI-207. The lamps shall be connected in parallel with one side grounded to the chassis of the unit and the other side being connected to a separate contact, in the electrical connector receptacle of the unit, for connection to an external +28 V input lead. The brilliance of the lamps will be controlled externally and the brilliance control shall not be part of the control unit. The arrangement of controls and all lettering and markings proposed for use on the front panel of control units shall be approved by the Bureau of Aeronautics prior to fabrication of any complete units.

3.2.10 Input Power Requirements - The equipment shall meet all applicable requirements of Specification MIL-E-7894 and shall give specified performance from the following power sources with characteristics as defined in Specification MIL-E-7894 having limits as specified hereinafter. The power required shall not exceed the specified amounts.

(1) AC Power (Single-Phase)

(a) 115-Volt Constant Frequency (Type I)

Operating Voltage Limits	108 to 121 volts
Operating Frequency Limits	380 to 420 cps
Reduced Performance Voltage Limits	102 to 108 and 121 to 124 volts
Power (1Ø)	37 V. A.

(2) DC Power 28-Volt System

Operating Voltage Limits	25 to 29 volts
Current	2.5 Amps

3.2.11 Input Power Connection - Input power connections shall be in accordance with MS 25245.

3.2.12 Overload Protection - Overload protection for the equipment shall be provided in the equipment. All parts and circuits of the equipment which are likely to carry an overload due to any failures or poor adjustments shall have suitable protective devices or shall be proportioned to withstand such overload without permanent damage to the equipment. The use of fuses and other protective devices shall be held to a minimum.

3.2.12.1 Undervoltage Protection - The equipment shall not be damaged by voltages below the minimum specified herein and shall automatically resume normal operation when the voltage returns within limits.

3.2.13 Standby Provisions - The equipment shall provide for a standby position wherein rated voltages shall be applied to all tube filaments, capstan drive motor, record amplifiers, blower motor, and any heaters necessary for low temperature operation.

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3.2.14 Nomenclature and Nameplates - Nomenclature assignment and nameplate approval for equipment identification shall be in accordance with Specification MIL-N-18307.

3.2.15 Interchangeability - The interchangeability requirements between like assemblies, subassemblies and replaceable parts as specified in Specification MIL-E-5400 shall be extended to require interchangeability of units of this equipment with the latest model of any previously produced equipment. Such interchangeability shall be measured against a model, manufacturing drawings, or other technical information as provided by the Bureau of Aeronautics for the purpose. In the event that requirements of Specification MIL-E-5400 conflict with the requirements for interchangeability, the interchangeability requirements shall govern.

3.2.16 Maintenance Provisions and Field Testing - Provisions for maintenance shall be as specified in Specification MIL-E-5400. Specific test points and test facilities shall be provided to the greatest extent practicable for ease of field testing and maintenance.

3.2.17 Radio Interference Control - The generation of radio interference by the equipment and the vulnerability of the equipment to radio interference shall be controlled within the limits of Specification MIL-I-6181.

3.3 Performance - The equipment shall provide recording facilities only (no playback) and shall be capable of making simultaneous recordings on each of two tracks in accordance with the following performance requirements:

3.3.1 Warm-Up Time - The warm-up time shall not exceed 5 minutes under standard conditions or 15 minutes under service conditions.

3.3.2 Recording Medium - The recording medium shall be magnetically coated tape of 0.25 inch width, using a mylar base. A nominal base tape thickness of 0.001 inch is required but the recorder shall also be capable of using mylar tape with a base thickness of 0.0015 inch.

3.3.3 Tape Speed - The tape speed shall be 7.5 inches per second.

3.3.3.1 Speed Regulation - Speed regulation shall not exceed ± 1% under standard conditions or ± 3% under service conditions.

3.3.4 Tape Breakage - The equipment shall not break tape under any normal sequence of operations except during the recording of the last foot of tape remaining on the supply reel. In the latter case, tape breakage not to exceed 1 foot in length fromthe end of the tape is permissible.

3.3.5 Unloaded Operation - No damage shall occur to the tape, except as is permissible under 3.3.4, or to any of the recorder parts should the recorder continue in operation after exhaustion of the tape from the supply reel.

3.3.6 End of Tape Indication - The equipment shall provide means for actuating an indicator light when approximately 5 minutes of recording time remain on the tape.

3.3.7 Standby-to-Record Time - Complete transfer from the standby mode to the record mode shall occur within 0.5 second after operation of the Standby-Record switch.

3.3.8 Input Levels - The equipment shall be capable of recording microphone or line inputs supplied at the levels indicated below:

Type	Impedance (ohms)	RMS Voltage Level	
		Max.	Min.
Microphone	100	0.5	0.016
Line	600	15.0	0.46

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3.3.9 Microphone Current - The equipment shall be capable of supplying 50 ± 7 ma of microphone current when operated in the record mode on either of the two tracks.

3.3.10 Other Performance Requirements - The following performance requirements are partially or entirely dependent upon the playback means. The AN/UNQ-7 Recorder-Reproducer Set, Sound, as specified under Specification MIL-R-21091 or a suitable equivalent will normally provide the required playback facilities. In the event of conflict between the requirements specified herein and the specification requirements governing the playback means, the requirements of this specification will apply.

3.3.10.1 Frequency Response - On both tracks, the frequency response between 20 and 10,000 cps shall be within ± 2 db of the response at 1000 cps. At temperature extremes of -55°C and +71°C, this tolerance may increase to not more than ± 3 db.

3.3.10.1.1 Equalization - Equalization circuitry shall be employed to provide approximately 7.5 db boost in the recording level at 10 kc on both tracks. Design of such record equalization circuits shall compensate for the response characteristic of the playback unit in order to obtain the required overall frequency response.

3.3.10.2 Distortion - The total harmonic distortion of recorded signals shall not exceed 2% at input signal frequencies between 20 and 10,000 cps. This requirement is overall and includes any distortion which may be contributed by the playback facilities. At temperature extremes of -55°C and +71°C, this tolerance may increase to not more than 4%.

3.3.10.2.1 Input Voltage Range - The distortion requirements shall apply over the input voltage ranges specified in 3.3.8.

3.3.10.3 Signal-to-Noise Ratio - The signal-to-noise ratio of signals recorded at minimum input level on either track shall not be less than 40 db.

3.3.10.4 Erasure - Erasure of the tape shall automatically occur during the recording process in such a manner that both tracks shall be erased prior to reaching the recording heads. The erased signal shall be not less than 45 db below the level recorded prior to erasure.

3.3.10.5 Isolation - The isolation between any audio circuits not directly interconnected, except through isolation networks, shall be at least 50 db.

3.3.10.6 Flutter - The amount of flutter and wow shall not exceed 0.5% rms during any portion of a tape recording when the recording is reproduced. (See 6.3)

3.3.10.7 Crosstalk - The crosstalk between tracks shall be at least 40 db down from the desired signal when both tracks are reproduced and compared.

3.3.10.8 Stability - The equipment shall be free of spontaneous changes in operating characteristics and shall show no evidence of spurious regeneration or oscillation.

3.3.10.9 Time Standard - The equipment shall be capable of generating a 1000 cps time standard within an accuracy of ± .01%. The capability of recording this signal on at least one track shall be required.

3.4 Detail Requirements

3.4.1 Recorder, Sound RO-28/UNH-6 - The sound recorder unit shall meet the following requirements.

3.4.1.1 Function - The sound recorder unit shall make recordings, during flight, of interphone conversations and radio audio output signals. Such recordings shall be made on either one or both tracks provided by the equipment. It is intended that the recorded tapes will be reproduced on Sound Recorder Reproducer, AN/UNQ-7, or equivalent.

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3.4.1.2 Form Factor and Mounting - All components of the sound recorder unit shall be enclosed in a single case of 4-7/8 inches width, 7-5/8 inches height, and 21-5/8 inches length. The unit shall be capable of being mounted in an aircraft by means of Mounting MT-(*)/UNH-6. The mounting base shall meet the applicable requirements of Specification MIL-C-172.

3.4.1.3 Weight with Mounting - The weight of the sound recorder unit shall not exceed 19.0 pounds and the mounting base therefor shall not exceed 4.0 pounds.

3.4.1.4 Contents of Unit - The unit shall contain the complete tape transport mechanism, electronic subassemblies, terminal strip board, connectors, and all other parts necessary to make the equipment function as a sound recorder. The chassis shall be mounted on rollers within the case in order to provide easy access for changing tape reels. A shatterproof glass window shall be provided in the case in order to permit observation of the tape transport mechanism and the Recording Time Indicators.

3.4.1.4.1 Input Terminal Board - An input terminal board shall be provided and shall contain suitable strapping arrangements whereby the input level requirements of each track may be altered to accommodate either microphone or line inputs (see 3.3.8). When the terminal board is strapped for a microphone input, the equipment shall be capable of supplying the required microphone current.

3.4.1.4.2 Meter Amplifier - A cathode follower type amplifier shall be provided for checking the record level on either track. The amplifier shall be designed to excite a VU meter.

3.4.1.4.3 Head Assembly - The head assembly shall be provided with the following facilities:

(1) ERASE - Full tape width erase

(2) RECORD - 2 tracks

(a) Single Track Width - 0.083 ± 0.003 inch

(b) Track Spacing - 0.084 ± 0.003 inch

(c) Record Gap Width - 0.001 inch

(d) Record Gap Scatter - 0.0004 inch

(e) Record Gap Azimuth - Fixed azimuth alignment. Azimuth alignment shall be within ± 2 minutes of arc measured from the perpendicular to the line of tape travel.

3.4.1.5 External Controls - The following front panel controls shall be provided on the sound recorder unit.

3.4.1.5.1 "ON-OFF, POWER" - The power switch shall be a toggle switch type and shall energize a power relay when operated to the "ON" position. The following standby conditions shall prevail with the "RECORD-STANDBY" switch in the "STANDBY" position.

(1) The capstan drive motor shall start and operate continuously. However, no tape motion shall occur.

(2) The filaments of all tubes in the sound recorder unit shall be energized.

(3) The power supply shall be energized and plate power shall be supplied to the record amplifiers.

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(4) The end of tape indication circuit shall be enabled but not energized until 5 minutes of tape recording time remain.

(5) The blower motor shall start and operate continuously.

3.4.1.5.2 "RECORD-STANDBY" - The "RECORD-STANDBY" switch shall be a toggle switch type and shall energize the record relay when operated to the "RECORD" position. The equipment shall then be ready to record information and the following conditions shall prevail in addition to those specified in 3.4.1.5.1.

(1) The capstan solenoid shall be energized and shall cause the capstan idler to press the tape against the rotating capstan. The tape shall then be transported from the supply reel to the take-up reel.

(2) The take-up motor shall be energized and the take-up reel shall wind the tape supplied from the capstan.

(3) The bias and erase oscillator shall be energized.

3.4.1.5.3 "CHANNEL SELECTOR" - The "CHANNEL SELECTOR" switch shall be a toggle switch type and shall effect transfer of the "RECORD LEVEL INDICATOR" between tracks. The two track positions on this switch shall be labelled "CHANNEL A" and "CHANNEL B".

3.4.1.5.4 "LEVEL" - Individual "LEVEL" controls on each track shall be provided for adjustment of the inputs to normal recording level. These controls shall be screwdriver types and each shall be covered by a springloaded access cover.

3.4.1.5.5 "RECORD LEVEL INDICATOR" - A "RECORD LEVEL INDICATOR" consisting of a miniature VU meter shall be provided for input level adjustment and monitoring purposes. Normal recording level shall be obtained when the meter needle swings to zero vu on signal peaks. The meter shall have a logarithmic scale with unit indications for the first positive three vu and indications at 0, -5, -10, and -20 vu.

3.4.1.5.6 "RELEASE-LOCK" - A lever shall be provided on the front panel to enable changing and/or installing reels. In the "RELEASE" position, it shall be possible to slide the sound recorder chassis out of the dust cover until it locks in the position which will permit easy removal or installation of both reels and tape, if any. After the sound recorder chassis is replaced within the dust cover, operation of the lever to the "LOCK" position shall prevent any further movement of the chassis on its rollers.

3.4.1.6 Internal Controls - The following screwdriver locking type adjustments shall be located internally on appropriate subassemblies of the sound recorder unit.

3.4.1.6.1 Meter Calibrate Adjust - A calibration adjustment shall be provided on each track to permit calibration of the "RECORD LEVEL INDICATOR".

3.4.1.6.2 Bias Adjust - Separate controls shall be provided on each track in order to adjust the bias supplied by the bias and erase oscillator.

3.4.1.7 Electrical Connections - All electrical connections to this unit shall be made through not more than two multi-contact AN connectors on the back of Mounting MT-(*)/UNH-6.

3.4.1.8 Recording Time Indicators - The tape transport shall contain elapsed and remaining time markings based on the use of 1800 feet of 1-mil mylar base tape.

3.4.2 Control, Recorder, C-(*)/UNH-6 - The conrol unit shall be of the console type in accordance with MIL-C-6781 and shall fulfill the following requirements:

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3.4.2.1 Function - This unit shall provide remote control of the AN/UNH-6 Sound Recorder Set and shall generate a 1000 cps time standard which may be recorded on one of the two tracks.

3.4.2.2 Form Factor - The dimension parallel to the mounting rails shall not exceed 2-5/8 inches while the overall depth measured from the rear of the front panel to the rear of the plug shall not exceed 4-1/2 inches.

3.4.2.3 Weight - The weight of this unit shall not exceed 2.7 pounds.

3.4.2.4 Contents of Unit - This control unit shall contain all circuits necessary to control primary power to the AN/UNH-6, to select "RECORD" or "STANDBY" modes of operation, and to generate and control the 1000 cps time standard.

3.4.2.5 External Controls - The following controls shall be located and suitably marked on the front panel:

3.4.2.5.1 "ON-OFF" Power - This switch shall be as specified in 3.4.1.5.1.

3.4.2.5.2 "RECORD-STANDBY" - This switch shall be as specified in 3.4.1.5.2.

3.4.2.5.3 "END OF TAPE" - An indicator light shall be provided and shall be lighted when approximately 5 minutes of recording time remain on the tape.

3.4.2.5.4 "TIME STANDARD"-The "TIME STANDARD" switch shall be a toggle type which shall enable a tuning fork transistorized oscillator.

3.4.2.6 Internal Controls - There shall be no internal adjustable controls.

3.4.2.7 Electrical Connections - All connections to this unit shall be made through one type AN plug mounted on the rear of the unit and extending through the dust cover.

3.4.3 Indicator, IP-(*)/UNH-6 - This indicator shall be of the console type in accordance with Specification MIL-C-6781 and shall fulfill the following requirements:

3.4.3.1 Function - This unit shall provide a remote indication of signal input levels to the AN/UNH-6 Sound Recorder Set.

3.4.3.2 Form Factor - The dimension parallel to the mounting rails shall not exceed 3 inches.

3.4.3.3 Weight - The weight of this unit shall not exceed 4 pounds.

3.4.3.4 Contents of Unit - This unit shall contain a minature cathode-ray tube, transistorized power supply, transistorized deflection amplifiers and all other circuits necessary to provide a visual monitor of the signal input to either track.

3.4.3.5 External Controls - The following controls shall be located and suitably marked on the front panel:

3.4.3.5.1 "CHANNEL SELECTOR" - The "CHANNEL SELECTOR" switch shall be a toggle switch type and shall effect transfer of the CRT monitor between tracks. The two track positions on this switch shall be labelled "CHANNEL A" and "CHANNEL B" and shall correspond to those on the RO-28/UNH-6 Sound Recorder Unit.

3.4.3.5.2 "PEAK LEVEL INDICATOR" - The face of the CRT shall be suitably etched and labelled to indicate the peak signal levels which correspond to the inputs at which normal record levels (see 6.2) were established in the RO-28/UNH-6 Sound Recorder Unit. The face of the CRT shall then provide a suitable visual presentation of the incoming signal and permit

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adjustment of peaks to the values required by the normal record levels established at the RO-28/UNH-6. Adjustment of signal peaks in this manner shall effected at the source or elsewhere external to the AN/UNH-6 Sound Recorder Set

3.4.3.6 Internal Controls - The following screwdriver locking controls shall be provided:

3.4.3.6.1 Centering Controls - Two controls shall be provided for centering the CRT picture; one horizontal, the other vertical.

3.4.3.6.2 Peak Level Calibrate - Two controls shall be provided for calibration adjustment. Each control shall establish the peak signal level between the etched lines on the face of the CRT for the corresponding position of the "CHANNEL SELECTOR" switch on this control unit.

3.4.3.7 Electrical Connections - All connections to this unit shall be made through one type AN plug mounted on the rear of the unit and extending through the dust cover.

3.4.3.8 Other Requirements - The face of the CRT tube shall be recessed to the maximum extent practicable with the size limitations specified herein but shall be not less than 0.250 inch from the rear of the front panel.

3.5 Workmanship - Workmanship, fabrication, and manufacturing processes shall conform to Specification MIL-E-5400.

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4. QUALITY ASSURANCE PROVISIONS

4.1 General - Equipments shall be subjected to the following tests to determine conformance with all the applicable requirements:

(1) Preproduction Tests

(2) Acceptance Tests

(3) Life Test

4.2 Preproduction Tests - Preproduction tests shall be made on one or more equipments representative of the production equipments to be supplied under the contract. Preproduction tests shall consist of the following:

(1) Contractor's Demonstration Tests

(2) Service Approval Tests

4.2.1 Contractor's Demonstration Tests - Contractor's demonstration tests shall be accomplished under the responsibility of the contractor and shall be conducted in accordance with the approved test procedure of 4.5. Data obtained by the contractor in conducting these tests shall be submitted to the Bureau of Aeronautics for review and approval prior to shipping the equipment to the specified destination for service approval tests. The Government Inspector and the Bureau of Aeronautics shall be advised when tests are to be conducted so that a representative may be designated to witness or supervise the tests when so desired. Contractors not having adequate facilities to conduct all required tests shall obtain the services of a commercial testing laboratory satisfactory to the Bureau of Aeronautics.

4.2.2 Service Approval Tests - The service approval model shall consist of one equipment representative of the production equipment to be supplied under the contract. This may be the same equipment that was subjected to the contractor's demonstration tests. This equipment shall be delivered to the destination specified by the Bureau of Aeronautics. When the tests of 4.2.1 are supervised, the Bureau of Aeronautics may, when conducting service approval tests, omit all tests which will duplicate tests previously conducted by the contractor; however, in conducting service approval tests, the Bureau of Aeronautics reserves the right to repeat any test previously conducted when deemed necessary.

4.2.2.1 Accessory Material - In addition to the complete equipment submitted for service approval tests, the contractor shall also submit the accessory material and design and test data specified in Specification MIL-E-5400. The design and test data shall include the data obtained by the contractor in testing to Specification MIL-T-5422. This information shall indicate the physical and electrical characteristics of the equipment and establish the equipment's compliance with applicable requirements.

4.2.3 Scope of Tests - Preproduction tests shall include all tests deemed necessary to determine that the equipment meets all the requirements of this specification and the contract. Preproduction tests shall include environmental tests in accordance with Specification MIL-T-5422 and interference tests in accordance with Specification MIL-I-6181. The sampling and testing of plastic lighting plates shall be in accordance with Specification MIL-P-7788.

4.2.4 Preproduction Approval - Approval of the preproduction model shall be by the Bureau of Aeronautics upon satisfactory completion of all tests. No production equipments shall be delivered prior to the approval of the preproduction model. Prefabrication of production equipment prior to the approval of the preproduction model is at the contractor's own risk. The approved preproduction model will be returned to the contractor for his use in the fabrication and testing of equipment to be submitted for acceptance. The preproduction model shall not be considered as one of the equipments under the contract; however, it may be reworked by the contractor and submitted for acceptance as a production equipment.

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4.3 Acceptance Tests - The contractor shall furnish all samples and shall be responsible for accomplishing the acceptance tests. All inspection and testing shall be under the supervision of the Government Inspector. Contractors not having testing facilities satisfactory to the Bureau of Aeronautics shall engage the service of a commercial testing laboratory acceptable to the Bureau of Aeronautics. The contractor shall furnish test reports showing quantitative results for all acceptance tests. Such reports shall be signed by an authorized representative of the contractor or laboratory as applicable. Acceptance or approval of material during the course of manufacture shall not be construed as a guarantee of the acceptance of the finished product. Acceptance tests shall consist of the following:

(1) Individual Tests

(2) Sampling Tests

(3) Special Tests

4.3.1 Individual Tests - Each equipment submitted for acceptance shall be subjected to the individual tests. These tests shall be adequate to determine compliance with the requirements of material, workmanship, operational adequacy and reliability. As a minimum, each equipment accepted shall have passed the following tests:

(1) Examination of Product

(2) Operational Test

(3) Reliability Test

4.3.1.1 Examination of Product - Each equipment shall be examined carefully to determine that the material and workmanship requirements of Specification MIL-E-5400 have been met.

4.3.1.2 Operational Test - Each equipment shall be operated long enough to permit the equipment temperature to stabilize and to check sufficient characteristics and record adequate data to assure satisfactory equipment operation.

4.3.1.3 Reliability Test - Each equipment shall be operated for a period of 6 hours without failure, under the conditions specified herein. A failure shall be defined as anything which causes malfunctioning of the equipment. Only those adjustments will be permitted which can be made by using such controls and adjustments accessible to the operator during the normal use of the equipment.

Temperature	Ambient room
Humidity	Ambient room
Vibration (10 minutes)	Any selected frequency within the range of 20 to 30 cps (excluding resonant points) and a minimum amplitude of ± 3 g's.

The equipment shall be vibrated (without vibration isolators) for a period of 10 minutes prior to the beginning of the 6-hour period of operation. Where feasible, the equipment shall be operated during this vibration period for the purpose of detecting flaws and imperfect workmanship. Operation within the specified limits of satisfactory performance is not necessarily required during the vibration period. The direction of vibration should be vertical to the normal mounting plane for 5 minutes and lateral to that plane for 5 minutes. Where it is not feasible to vibrate the equipment in 2 directions the vertical direction shall be used. During the 6-hour period of operation following the 10-minute vibration period, the equipment shall be mechanically cycled periodially through its various phases of operation. Should a failure occur, it should be repaired and the test started over, except the 10-minute vibration period need not be repeated when it is certain the failure was not a result of the vibration. Should repetitive failures occur, corrective action shall be taken to eliminate this defect from future equipment. A record shall be kept of all failures. The 6-hour period specified above may be composed of two 3-hour periods to conform with standard working hours.

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4.3.2 Sampling Tests - Sampling tests shall be conducted on complete equipments which have successfully passed the individual tests. The following equipments shall be selected by the Government Inspector for sampling tests.

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 10	1
11 to 100	2
101 to 200	1
Over 200	1 for each additional 200 or fraction thereof

4.3.2.1 Scope of Tests - As a minimum, each equipment selected for sampling tests shall be subjected to the following tests:

(1) Complete operational test at ambient room conditions and nominal input voltages, making all necessary measurements to assure that all applicable specification requirements have been met.

(2) Repeat the test of (1) above using the highest and lowest specified input voltages and frequencies.

(3) Reliability test specified in 4.3.1.3, except that the test duration shall be 30 hours.

4.3.3 Special Tests - Special tests shall be conducted on a quantity of equipments for the purpose of checking the effect of any design or material change on the performance of the equipment and to assure adequate quality control. The equipment selected for special tests may be selected from equipments previously subjected to the sampling test.

4.3.3.1 Special Tests Schedule - Selection of equipments for special tests shall be made as follows:

(1) On an early production equipment

(2) On an equipment midway of the production run.

(3) On an early equipment after an engineering change.

(4) Whenever failure reports or other information indicate additional equipments should be tested. (This will be determined by the Bureau of Aeronautics.)

Exceptions from the above may be necessary when the number of equipments procured under contract or other factors affecting performance may require a variation to assure satisfactory quality control.

4.3.3.2 Scope of Tests - Special tests shall consist of such tests as approved by the Bureau of Aeronautics. Test procedures previously approved for the preproduction and sampling tests shall be used where applicable. When not applicable, the contractor shall prepare a test procedure and submit it to the Bureau of Aeronautics for approval prior to conducting the tests. As a minimum, the following tests shall be made, except that for those equipments selected as a result of design changes and failure reports, only such tests required to check the characteristics in question need be conducted.

(1) Temperature - Altitude Tests

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(2) Vibration Test (90 minutes cycling - in each of 3 planes)

(3) Humidity Tests

4.3.4 Equipment Failure - Should a failure occur during either the sampling or special tests, the following action shall be taken:

(1) Determine the cause of failure.

(2) Determine if the failure is an isolated case or design defect.

(3) Submit to the Bureau of Aeronautics for approval, proposed corrective action intended to reduce the possibility of the same failure(s) occurring in future tests.

(4) Where practical, include a test in the individual test to check all equipment for this requirement until reasonable assurance is obtained that the defect has been satisfactorily corrected.

4.4 Life Test - A 300-hour life test shall be conducted at the contractor's plant on an early equipment representative of the current production. The life tests shall be performed under the conditions specified in 4.4.1. The life test sample shall be selected by the Government Inspector in accordance with the following: (Equipments which have successfully passed the sampling tests or special tests may be selected for life tests.)

Quantity of Equipments Offered for Acceptance	Quantity to be Tested
First 25	1
26 to 200	1
201 to 500	1
Over 500	1 for each additional 500 or fraction thereof

4.4.1 Test Conditions - The life test shall be conducted under the following simulated service conditions:

Temperature	Normal room temperature
Altitude	Normal room
Humidity	Room ambient
A. C. Voltage	115 ± 5 volts (at lowest applicable frequency)
D. C. Voltage	26.5 ± 2.5 volts

4.4.2 Test Periods - The test may be run continuously or intermittently. Any period of operation shall be of sufficient duration to permit the equipment temperature to stabilize. Periodically, the equipment shall be turned on and off several times and put through its various phases of operation.

4.4.3 Performance Check - At approximately 8-hour intervals during the test, a limited performance check shall be made. The performance check proposed by the contractor shall be subject to approval by the Bureau of Aeronautics.

4.4.4 Test Data - The contractor shall keep a daily record of the performance of the equipment, making particular note of any deficiencies or failures. In the event of part failures, the defective part shall be replaced and the operation resumed for the balance of the test period. A record shall be kept of all failures throughout the test, including all transistor or tube failures. This record shall indicate the following:

(1) Part type number

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(2) The circuit reference symbol-number

(3) The part function

(4) Name of the manufacturer

(5) Nature of the failure

(6) The number of hours which the part operated prior to failure

4.4.4.1 Failure Report - In the event of a failure, the Government Inspector shall be notified immediately. A report shall be submitted to the Bureau of Aeronautics upon completion of the test. In this report, the contractor shall propose suitable and adequate design or material corrections for all failures which occurred. The Bureau of Aeronautics will review such proposals and determine whether they are acceptable.

4.4.5 Reconditioning of Life Test Samples - An equipment which has been subjected to the life tests shall be reconditioned as follows:

(1) On completion of the life test, the equipment shall be reworked by the contractor by replacing all tubes and "wear" items. The "wear" items shall be determined by agreement between the contractor and the Bureau of Aeronautics.

(2) After reworking, the contractor shall resubmit the equipment for acceptance.

4.5 Test Procedures - The procedures used for conducting preproduction tests, acceptance tests and life tests shall be prepared by the contractor and submitted to the Bureau of Aeronautics for review and approval. The right is reserved by the Bureau of Aeronautics or the Government Inspector to modify the tests or require any additional tests deemed necessary to determine compliance with the requirements of this specification or the contract. Specification MIL-T-18303 shall be used as a guide for preparation of test procedures. When approved procedures are available from previous contracts the Bureau of Aeronautics will furnish the procedures to be used for testing.

4.6 Presubmission Testing - No item, part or complete equipment shall be submitted by the contractor until it has been previously tested and inspected by the contractor and found to comply, to the best of his knowledge and belief, with all applicable requirements.

4.7 Rejection and Retest - Equipment which has been rejected may be reworked or have parts replaced to correct the defects and resubmitted for acceptance. Before resubmitting, full particulars concerning previous rejection and the action taken to correct the defects found in the original shall be furnished the Government Inspector. Units rejected after retest shall not be resubmitted without the specific approval of the procuring agency.

4.8 Equipment Changes - After approval of the preproduction model, no changes which affect weight, installation interchangeability and interchangeability of maintenance parts or assemblies shall be incorporated in the equipment unless approved by the Bureau of Aeronautics. To obtain Bureau of Aeronautics approval, the contractor shall submit an Engineering Change Proposal (ECP) giving the necessary information concerning the change. This shall apply for either a Bureau of Aeronautics' recommended change or a contractor's proposed change. When required, the contractor shall also prepare an Electronic Material Change (EMC) and an Electronic Material Bulletin (EMB) to cover the approved change. ECP's, EMC's and EMB's shall be prepared in accordance with SAR-300.

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5. PREPARATION FOR DELIVERY

5.1 General - All major units and parts of the equipment shall be preserved, packaged, packed and marked for the level of shipment specified in the contract or order in accordance with Specification MIL-P-17555.

6. NOTES

6.1 Test Values - Normal and limiting values of performance data shall be determined at input voltages of 27.5 ± 0.5 V DC and 115 ± 1.0 V AC. These data are to be used in testing the equipment at installation points for compliance with minimum acceptable standards of performance.

6.2 The normal record level, as referred to herein, is that level of magnetization of the recording medium at which the rms total of the distortion products of a 400-cycle tone, measured at the output of the playback unit, reaches 1% of the rms value of the 400-cycle tone due to saturation of the recording medium.

6.3 Flutter - In order to establish a uniform method, flutter shall be measured as follows:

(1) A 3000-cycle tone shall be recorded on the tape.

(2) The tape record shall be reproduced on an AN/UNQ-7 Sound Recorder Reproducer or equivalent.

(3) The reproduced signal shall be fed into a suitable flutter bridge, and flutter content measured on a calibrated bridge output meter. The response of the flutter bridge shall not vary more than 6 db between 0.5 and 300 cps.

NOTICE - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished or in any way supplied the said drawings, specifications or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

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